9. If $222^2 + 2 \times 222 \times 178 + 178^2 = p$, then $\sqrt{p}$ is

1) $25^2 - 15^2$ 2) $20^2 + 20^2$ 3) $50^2 - 10^2$ 4) $16^2 - 12^2$

10. If $P = 2x^2 + 3xy + 5y^2$ and $Q = 14x^2 + 37xy + 20y^2$, then P + Q is

1) $(4x - 5y)^2$ 2) $(5x + 4y)^2$ 3) $(5x - 4y)^2$ 4) $(4x + 5y)^2$

11. If $(a + 5)(a - 5)(a^2 + 25)$ is simplified, then the result is

1) $a^4 - 125$ 2) $a^4 - 25$ 3) $a^4 - 625$ 4) $a^5 - 3125$

12. If $13x = 58^2 - 45^2$, $24y = 99^2 - 87^2$, then x + y is

1) 227 2) 276 3) 196 4) 274

13. If $\dfrac{23.71 \times 23.71 - 16.29 \times 16.29}{(23.71 + 16.29)^2}$ is simplified, then the result is

1) $\dfrac{-742}{4000}$ 2) $\dfrac{742}{4000}$ 3) $\dfrac{4000}{742}$ 4) $\dfrac{-4000}{742}$

14. For the product $\left(\frac{3}{5}p + \frac{1}{3}\right)\left(\frac{3}{5}p - \frac{1}{3}\right)$ is the value obtained by using the identity is

1) $\dfrac{9p^2}{25} + \dfrac{1}{9}$ 2) $\dfrac{1}{9} - \dfrac{9p^2}{25}$ 3) $\dfrac{9p^2}{25} - \dfrac{1}{9}$ 4) $\dfrac{3}{5}p - \dfrac{1}{3}$

15. Using the identity $(a + 2)(a - 2) = a^2 - b^2$, the value obtained from the product $\left(\frac{2}{5} + x\right)\left(\frac{2}{5} - x\right)\left(\frac{4}{25} + x^2\right)$ is

1) $\dfrac{16}{625} - x^4$ 2) $\dfrac{16}{625} + x^4$ 3) $x^4 - \dfrac{16}{625}$ 4) $x^4 + \dfrac{16}{25}$

16. The expansion $4a^2 + \frac{9}{25} + \frac{12}{5}a$ as a perfect square is

1) $\left(2a + \dfrac{3}{5}\right)^2$ 2) $\left(2a - \dfrac{3}{5}\right)^2$ 3) $\left(\dfrac{3}{5} - 2a\right)^2$ 4) $\left(2a^2 - \dfrac{3}{5}\right)^2$

17. The missing term in the perfect square _____ + 90xy + $81y^2$ is

1) 9y 2) 90x 3) 5x 4) $25x^2$

18. With out actual multiplication, the value of $79.01 \times 79.01 + 2 \times 79.01 \times 20.99 + 20.99 \times 20.99$ is

1) 10.009 2) 1000.05 3) 10,000 4) 10007

19. Simplified form of $\dfrac{5718 \times 5718 - 4135 \times 4135}{5718 + 4135}$ is

1) 1683 2) 1583 3) 1783 4) 1563

5. If pair of linear equations $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are intersecting lines.

1) $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ 2) $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$ 3) $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ 4)None

6. The system of equations 3x + 2y + 5 = 0 and 9x + 6y + 4 = 0 has

1)Unique solution 2)Infinite solutions 3)No solution 4)None

7. For what value of ' p ' the system of equations 2x -py = 0 and 3x + 4y = 0 have non-zero solution?

1)p = - 6 2)p = $\frac{-8}{3}$ 3)p = $\frac{-2}{3}$ 4)p = 4

8. The system of equations 5x + 2y = 16 , $\frac{15}{2}x + 3y = 24$

1) has a unique solution 2) has no solution

3) has infinitely many solutions 4) are inconsistent.

9. If the given system of the equations kx + 3y – (k – 3) = 0,12x + ky – k = 0

has infinitely many solutions if k =

1) 3 2) 2 3) 6 4) 7

10. For what values of ' k ' ,the system of equations kx - y = 2 , 6x- 2y = 3 has infinite solutions.

1) k = 3 2)k = 6 3)k = 4 4)Does not exist

11. If 3x – 5 < 10 < 2x + 5, then range of x is

1) $\frac{3}{2} < x < 3$ 2) $\frac{5}{2} < x < 4$ 3) $\frac{5}{2} < x < 5$ 4) None

12. Given $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$ where u = 30 v = 20 then f = _____.

1) 25 2) 20 3) 15 4) 12

9. The two diagonals are not necessarily equal in a

1)Rectangle 2square 3)rhombus 4)Isosceles trapezium

10. The lengths of the diagonals of a rhombus are 16 cm and 12 cm.

The length of each side of the rhombus is

1)8 cm 2)9 cm 3)10 cm 4)12 cm

11. Two sides of a parallelogram are in the ratio 5:4. If its perimeter is 54 cm, the length of the larger side is :

1)9 cm 2)11 cm 3)12 cm 4)15 cm

12. The bisectors of any two adjacent angles of a parallelogram intersect at

1)30^0 2)45^0 3)60^0 4)90^0

13. If an angle of a parallelogram is two-third of its adjacent angle, the smallest angle of the parallelogram is

1)54^0 2)72^0 3)81^0 4)108^0

14. Two adjacent angles of a parallelogram are $(2x+25)^0$ and $(3x-5)^0$. The value of x is

1)28 2)32 3)36 4)42

15. Any two adjacent angles of a parallelogram are:

1)Complementary 2)Supplementary

3)Equal 4)all of these

16. A quadrilateral having exactly one pair of parallel sides is a:

1)Square 2)rhombus 3)Trapezium 4)rectangle

20. If $64a^2+36b^2=400$, ab = 4, then 8a + 6b is

1) 26 2) 28 3) 22 4) 24

21. If $9x^2+48x+P$ to be a perfect square, then the value of P is

1) 81 2) 64 3) 36 4) 16

22. If $(9a+7b)^2+(9a-7b)^2$ is simplified , then the resultant is

1) $162a^2+98b^2$ 2) $126a^2+102b^2$ 3) $162a^2-126b^2$ 4) $126a^2-162b^2$

23. If $x-\frac{1}{x}=\frac{8}{3}$, then $x+\frac{1}{x}$ =

1) $\frac{10}{3}$ 2) $\frac{7}{3}$ 3) $\frac{5}{3}$ 4) $\frac{8}{3}$

24. If $x+\frac{1}{x}=10$, then $x-\frac{1}{x}$ =

1) $\pm4\sqrt{6}$ 2) $\pm2\sqrt{6}$ 3) $\pm3\sqrt{6}$ 4) $\pm5\sqrt{6}$

25. If $k^2 - 3k + 1 = 0$, then $k+\frac{1}{k}$ =

1) 3 2) 9 3) 27 4) 81

26. If $x-\frac{1}{x}=5$, then $x^3-\frac{1}{x^3}$ =

1) 100 2) 125 3) 110 4) 145

27. If x + y = 8 ,xy = 12 ,then $x^4+y^4=$

1) 1012 2) 1112 3) 1212 4) 1312

28. If $z-\frac{1}{z}=4$,then $z^2+\frac{1}{z^2}=$

1) 18 2) $8\sqrt{5}$ 3) $4\sqrt{5}$ 4) 8

29. If a + b + c = 0 then $a^3 + b^3 + c^3$ =

1) 3 2) 3abc 3) abc 4) 9abc

30. If f (x , y, z) = $x^3 + y^3 + z^3 - 3xyz$ then f(124 , 213 , -337)

1)0 2)12345 3)1 4)234567

ASSIGNMENT -KEY

01. 3	**02.** 3	**03.** 1	**04.** 2	**05.** 3	**06.** 1
07. 3	**08.** 1	**09.** 4	**10.** 4	**11.** 3	**12.** 2
13. 2	**14.** 3	**15.** 1	**16.** 1	**17.** 4	**18.** 3
19. 2	**20. 2**	**21.** 2	**22.** 1	**23.** 4	**24.** 1
25. 1	**26.** 3	**27.** 4	**28.** 1	**29.** 2	**30.** 1

13. If $4 \le x \le 9$ and $3 \le y \le 5$, find minimum value of $\frac{2y}{3+x}$.

1) 2/5 2) 2 3) 1/5 4) 1/2

14. If | x - 3 | = 2 then x =

1)2,4 2)1,5 3)1,6 4)5,4

15. If | x - 3 | < 2 then 'x'lies in the interval

1)(1,5) 2)(1,3) 3)(2,3) 4)(2,5)

15. If | x - 3 | >2 then

1) x < 1 or x > 5 2)x < 1 or x > 4 3)x < 3 or x > 5 4)x < 2 or x > 7

16. If $|3x-2| \le 5$ then x belongs to

1)[-1,7/5] 2)[-1,7/3] 3)[1,7/5] 4)[-1,7]

17. If $2 \le x < 4$, where 'x' is a natural number then the maximum value of x is

1)1 2)2 3)4 4)3

18. If $\frac{2}{3x+2y}+\frac{3}{3x-2y}=\frac{17}{5}$, $\frac{5}{3x+2y}+\frac{1}{3x-2y}=2$, then x + y =

1) 2 2) 3 3) 4 4) – 2

19. If the variable in the inequation $5x + 2 \le 12$ takes the values from {–4, –3, –2, –1, 0, 1, 2, 3, 4}, then the suitable values of variable x is

1) {– 4, – 3, – 2, – 1, 0, 1, 3} 2) {– 4, – 3, – 2, – 1, 0, 1, 2}
3) {– 3, – 2, – 1, 0, 1, 2, 3} 4) {– 2, – 1, 0, 1, 2, 3, 4}

20. If $3x + 4 \ge -14$, then the domain of the inequation is

1) $x \le -6$ 2) $x \ge -6$ 3) $x \le -9$ 4) $x \le -14$

21. If $\frac{3x}{5}+2 \le 17$, then the domain of the inequation is

1) x > 25 2) $x \ge 26$ 3) $x \ge 25$ 4) $x \le 25$

18. On substracting $\frac{3}{2}x^2y+\frac{4}{5}y-\frac{1}{3}x^2yz$ from $\frac{12}{5}x^2yz-\frac{3}{5}xyz+\frac{2}{3}x^2y$, the result is equal to

1) $\frac{41}{15}x^2yz-\frac{5}{6}x^2y+\frac{3}{5}xyz+\frac{4}{5}y$
2) $\frac{41}{15}x^2yz-\frac{5}{6}x^2y-\frac{3}{5}xyz+\frac{4}{5}y$
3) $\frac{41}{15}x^2yz-\frac{5}{6}x^2y-\frac{3}{5}xyz-\frac{4}{5}y$
4) $\frac{41}{15}x^2yz+\frac{5}{6}x^2y-\frac{3}{5}xyz+\frac{4}{5}y$

19. Subtracting $\frac{9}{2}+\frac{x}{2}+\frac{3}{5}x^2+\frac{7}{4}x^3$ from $\frac{7}{2}-\frac{x}{3}-\frac{x^2}{5}$, the result is equal to

1) $-1-\frac{5}{6}x-\frac{4}{5}x^2-\frac{7}{4}x^3$
2) $1-\frac{5}{6}x+\frac{4}{5}x^2+\frac{7}{4}x^3$
3) $1-\frac{5}{6}x+\frac{4}{5}x^2-\frac{7}{4}x^3$
4) $1-\frac{4}{5}x^2+\frac{7}{4}x^3-\frac{5}{6}x$

20. What must be subtracted from $x^4 + 2x^2 - 3x + 7$ to get $x^3 + x^2 + x - 1$ is

1) $x^4 - x^3 + x^2 - 4x + 8$
2) $x^3 + x^2 - 4x + 8$
3) $x^4 - x^3 + x^2 + 4x - 8$
4) $x^4 - x^3 - x^2\ \ 4x - 8$

21. Subtract $x^3 - xy^2 + 5x^2y - y^3$ from $-y^3 - 6x^2y - xy^2 + x^3$ is

1) $2y^3 - 8x^2y + 3xy^2 - 2x^3$
2) $2x^3 - 2xy^2 - x^2y - 2y^3$
3) $-11x^2y$
4) none of these

22. The value of $(5a^6) \times (-10ab^2) \times (-2.1\ a^2b^3)$ at a = 1 and b = $\frac{1}{2}$ is

1) $\frac{102}{35}$
2) $\frac{103}{32}$
3) $\frac{-105}{32}$
4) $\frac{105}{32}$

23. The product value of $(x^2\ y)^4 \times (2xy^4) \times (-3xy)$ at x = 2 and y = –1 is

1) 6155
2) 5144
3) 2174
4) 6144

24. The product of $100x \times (0.01\ x^4 - 0.01\ x^2)$ is

1) $0.01\ x^4 - x^3$
2) $0.5\ x^5 - x^3$
3) $x^5 - x^3$
4) $x^3 - x^5$

25. On multiplication of $\left(3x-\frac{4}{5}y^2x\right)$ by $\frac{1}{2}xy$ the result is

1) $\frac{3}{2}x^2y + \frac{5}{2}x^2y^3$
2) $\frac{2}{3}x^2y - \frac{5}{2}x^2y^3$
3) $\frac{3}{2}x^2y - \frac{2}{5}x^2y^3$
4) $\frac{5}{2}x^2y + \frac{3}{2}yx^2$

26. The product of $\left(\frac{1}{5}x-\frac{1}{4}y\right)$ and $(5x^2 - 4y^2)$ is

1) $x^3 + \frac{5}{4}xy^2 + \frac{5}{4}x^2y + y^3$
2) $x^3 + \frac{4}{5}xy^2 - \frac{5}{4}x^2y + y^3$

6. A is a 3-digit number below 200. It has no single-digit factor except 1. Find the number given that is not a perfect square.
   1) 187 2) 143 3) Either (1) or (2)
   4) Both (1) and (2)
7. Find the number n given that has the first ten natural numbers as its factors and is the smallest 5 digit number
   1) 12,520 2) 10,080 3) 36,280 4) 10,040
8. The smallest number that must be multiplied by 450 to make it a perfect cube is
   1) 2 2) 3 3) 6 4) 5
9. If $2m + 5n$ and $3m + 4n$ are two distinct prime numbers, then the numbers whose G.C.D. is 50 is
   1) $50m + 150n$ and $150m + 20n$ 2) $100m + 250n$ and $150m + 200n$
   3) $100m + 200n$ and $250m + 300n$ 4) $50m + 250n$ and $150m + 250n$
10. When a number is divided, the divisor is equal to three times the remainder or equal to the square of the remainder. If ten times the divisor is more than the quotient by a value equal to the remainder, then what is the dividend ?
    1) 857 2) 976 3) 786 4) Cannot be determined
11. When a number is divided by 9 the remainder is 7. If twice the number is divided by 9, then the remainder is
    1) 9 2) 7 3) 4 4) 5
12. Two positive integers are in the ratio of 3 : 4. Their HCF and LCM are in the ratio of 1 : 12. If the difference between their LCM and their HCF is 44, then sum of the numbers are
    1) 28 2) 48 3) 56 4) 120
13. For what values of k and m is $(19^k - 3^k)(18^m - 11^m)$ divisible by 112. $(k, m \in N)$
    1) When k is odd, m is even 2) When k is even m is odd
    3) For all values of k and m 4) cannot be determined
14. The number of prime numbers of the form $\frac{n(n+1)}{2}$, where 'n' is a natural number is
    1) n 2) 1 3) n – 1 4) 2
15. Which of the following are correct
    i) $x^n + y^n$ is divisible by x + y, only if n is an odd number
    ii) $x^n - y^n$ is divisible by x – y, only if n is an even number
    iii) $x^n - y^n$ is divisible by x + y, only if n is an even number
    iv) $x^n + y^n$ is not divisible by either x + y or x – y, if n is an even number
    1) (i) and (ii) 2) (ii), (iii)
    3) (i), (iii) and (iv) 4) All are correct

ASSIGNMENT

1. If $49(a^2 - b^2) + 28(a - b)^2 = A$ is added to $B = 16(a + b)^2 - 25(a^2 - b^2)$, then factors of $A + B - 44ab + 48b^2$ is
   1) $68(a^2 + ab + b^2)$ 2) $68(b^2 - ab - a^2)$
   3) $68(a^2 - ab + b^2)$ 4) $68(b^2 + ab - a^2)$
2. Factorize: $x^2y^2 + xyz + xy + z$
   1) $(xy-z)(xy + 1)$ 2) $(xy + z)(xy - 1)$
   3) $(xy - z)(xy - 1)$ 4) $(xy + z)(xy + 1)$
3. Factorize: $2mx - 2my - nx + ny$
   1) $(x-y)(2m + n)$ 2) $(x+y)(2m + n)$
   3) $(x-y)(2m - n)$ 4) $(x+y)(2m - n)$
4. Factorize: $2x^2 - 5x - 2xy + 5y$
   1) $(2x + 5)(x - y)$ 2) $(2x - 5)(x - y)$ 3) $(2x + 5)(x + y)$ 4) $(2x - 5)(x + y)$
5. Factorize: $ax - by + by + cy - cy - ay$
   1) $a(x - y)$ 2) $a(y - x)$ 3) $x(y - a)$ 4) $a(x + y)$
6. Factorize: $x^4 + x^3 + 5x + 5$
   1) $(x^3 - 5)(x - 1)$ 2) $(x^3 + 5)(x - 1)$ 3) $(x - 5)(x^3 + 1)$ 4) $(x^3 + 5)(x + 1)$
7. $\frac{1}{6}a^2 - \frac{8}{24}ab + \frac{1}{4}b^2$ is factorized and multiplied by $\frac{24}{7}(3a - 4b)$, then the result is
   1) $\frac{2}{7}(3a - 2b)(2a^2 - 4ab - 3b^2)$ 2) $\frac{2}{7}(3a - 4b)(2a^2 - 4ab + 3b^2)$
   3) $\frac{2}{7}(3a - 4b)(2a^2 + 4ab + 3b^2)$ 4) $\frac{2}{7}(3a + 4b)(2a^2 + 4ab + 3b^2)$
8. If $16(4a - 3b)^2 + 4(4a - 3b)$ is factorised, then its factors are
   1) $4(4a - 3b)(16a - 12b - 4)$ 2) $4(4a - 3b)(16a + 12b + 4)$
   3) $4(4a + 3b)(16a - 12b + 4)$ 4) $4(4a - 3b)(16a - 12b + 1)$
9. If $ax^2 + by^2 + bx^2 + ay^2$ is factorised, then its factors are
   1) $(x^2 + y^2)(a + b)$ 2) $(x^2 - y^2)(a - b)$ 3) $(x^2 - y^2)(a + b)$ 4) $(x^2 + y^2)(a - b)$
10. If $pq(x^2 + y^2) + xy(p^2 + q^2)$ is factorised, then the factors are
    1) $(py + qx)(px + qy)$ 2) $(px + qy)(p^2x + q^2y)$
    3) $(px + qy)(px + qy)$ 4) $(px^2 + qy^2)(qx^2 + py^2)$
11. If $6ab - b^2 + 12ac - 2bc$ is factorised, then its factors are
    1) $(b - 7c)(6a + b)$ 2) $(b + 2c)(6a - b)$ 3) $(b - 2c)(6a - b)$ 4) $(b + 2c)(6a + b)$

7. If $x^{\frac{1}{4}} = 4, y^{\frac{1}{2}} = 3$, then $\left(x^{\frac{3}{4}}\right) + \left(y^{\frac{5}{2}}\right) =$

1) 309 2) 306 3) 307 4) 310

8. If $p = \dfrac{125^{\frac{7}{3}} - 125^{\frac{5}{3}}}{125^{\frac{2}{3}}}$ and $y^{y+1} = 81$, then $P \times y^2$ is

1) $3^2 \times 10^4$ 2) $3^3 \times 10^2$ 3) $3^2 \times 10^3$ 4) $3^3 \times 10^3$

9. If $\sqrt[5]{\sqrt[4]{a}} = 2, \sqrt[2]{\sqrt[5]{b}} = 4,$ then $\sqrt[40]{ab}$ is

1) 4 2) 8 3) 16 4) 2

10. If $\sqrt[3]{\dfrac{a}{b}} = \dfrac{2}{3}, \sqrt[4]{\dfrac{x}{y}} = \dfrac{4}{3}$, then the value of $\sqrt[14]{\dfrac{ax \times 2^3}{by \times 3^7}}$ is

1) $\dfrac{2}{3}$ 2) $\dfrac{8}{27}$ 3) $\dfrac{16}{81}$ 4) $\dfrac{32}{243}$

11. If $\sqrt[2]{a} = x, \sqrt[3]{b} = y$ where $a + b = 9$, $a = 2 \times b$ then

1) $x = y$ 2) $x > y$ 3) $x < y$ 4) $x \leq y$

12. If $x = \sqrt[6]{a}, y = \sqrt[3]{b+1}, z = \sqrt[4]{c+4}$ are three radicals where $\dfrac{a}{2} = \dfrac{b+1}{3} = \dfrac{c+4}{4} = 5$ then they are written in descending order that sequence is

1) y, x, z 2) y, z, x 3) x, y, z 4) x, z, y

13. If $x = 2y = 4z = 8p = 64$ where $b = \sqrt[12]{x}, c = \sqrt[4]{y}, d = \sqrt[3]{z}, e = \sqrt{a}$ in ascending order is

1) b, c, d, e 2) b, c, e, d 3) d, b, c, e 4) a, c, d, b

14. If $\sqrt[m]{a^x}, \sqrt[n]{b^y}, \sqrt[p]{c^z}, \sqrt[r]{f^q}$ are radicals where a = 3, b = 9, c = 27, f = 81, x = y + 1 = z + 2 = q + 3 = 5, m = 2, n = 4, p = 8, r = 16, then descending order of the radicals is

1) $\sqrt[16]{3^{40}}, \sqrt[16]{3^{32}}, \sqrt[16]{3^{18}}, \sqrt[16]{3^{8}}$ 2) $\sqrt[16]{3^{40}}, \sqrt[16]{3^{8}}, \sqrt[16]{3^{32}}, \sqrt[16]{3^{18}}$

3) $\sqrt[16]{3^{18}}, \sqrt[16]{3^{8}}, \sqrt[16]{3^{40}}, \sqrt[16]{3^{32}}$ 4) $\sqrt[16]{3^{32}}, \sqrt[16]{3^{40}}, \sqrt[16]{3^{8}}, \sqrt[16]{3^{18}}$

15. If $\sqrt[3]{2}, \sqrt[4]{3}, \sqrt[2]{5}$ are arranged in ascending order, then the correct order is

1) $\sqrt[4]{3}, \sqrt[3]{2}, \sqrt[2]{5}$ 2) $\sqrt[2]{5}, \sqrt[4]{3}, \sqrt[3]{2}$ 3) $\sqrt[3]{2}, \sqrt[2]{5}, \sqrt[4]{3}$ 4) $\sqrt[3]{2}, \sqrt[4]{3}, \sqrt[2]{5}$

16. The value of $\sqrt[4]{81} - 8\sqrt[3]{216} + 15\sqrt[5]{32} + \sqrt{225}$ is

1) 1 2) 2 3) 3 4) 0

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CONTENTS

S.NO.	TOPIC	PAGE NO.
1.	Number System	5-27
2.	Law of Indices & Fractions	28-36
3.	Algebraic Expressions	37-46
4.	Factorisation	47-55
5.	Sets	56-61
6.	Linear Equations & Inequations	62-80
7.	Algebra Formulae	81-88
8.	Plane Geometry	89-106
9.	Mensuration	107-123